

Identifying a Drilling State Machine

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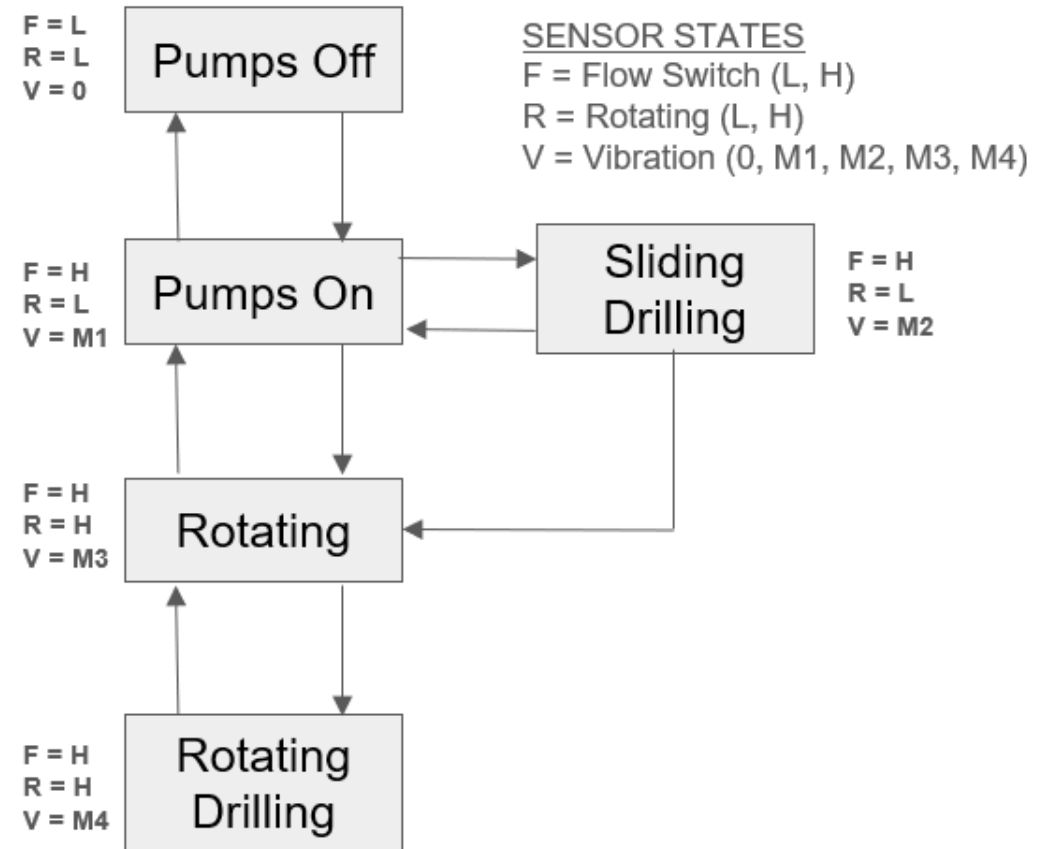
For Deere Development Company, LLC.

June 23, 2025

The Problem

- Software on drilling tools can be optimized by knowing what state the tool is in
- Deere Development Company (DDC) has outlined the Drilling State Machine seen here to fulfill their software needs
- Indicating Drilling Parameters
 - Flow
 - Rotation
 - Vibration
 - Weight on Bit (WOB)

DRILLING STATE MACHINE



Related
Previous
Work:
Supervised

Method	Random Forest (RF)
Industries	• Oil & Gas
Applications	• Geological Formation Prediction
Data	• Sensor Data

Related
Previous
Work:
Unsupervised

Method	Hidden Markov Model (HMM)	Gaussian Mixture Model (GMM)	Hierarchical Density-Based Spatial Clustering for Applications with Noise (HDBSCAN)
Industries	Oil & Gas	- Oil & Gas - Statistical Research	- Manufacturing - Natural Language Processing
Applications	Pump Jack Performance Analysis	- Pump Jack Performance Analysis - Oil Spill Image Classification	- Robotic Welding Monitoring - Record Classification
Data	- Large dataset - Time-series - Sensor Data	- Large dataset - Time-series - Sensor Data	- Real Time - Sensor Data

Well	Run	Date In	Date Out	MD In	MD Out
Flybar 1 WB	Run 7	12/2/2024	12/4/2024	10582	13222
	Run 8	12/4/2024	12/8/2024	13222	17880
Flybar 1 WC	Run 6	12/9/2024	12/10/2024	10175	10287
	Run 7	12/10/2024	12/11/2024	10287	10713
	Run 8	12/11/2024	12/12/2024	10713	12088
Flybar 2 WC	Run 5	11/23/2024	11/28/2024	10960	16782

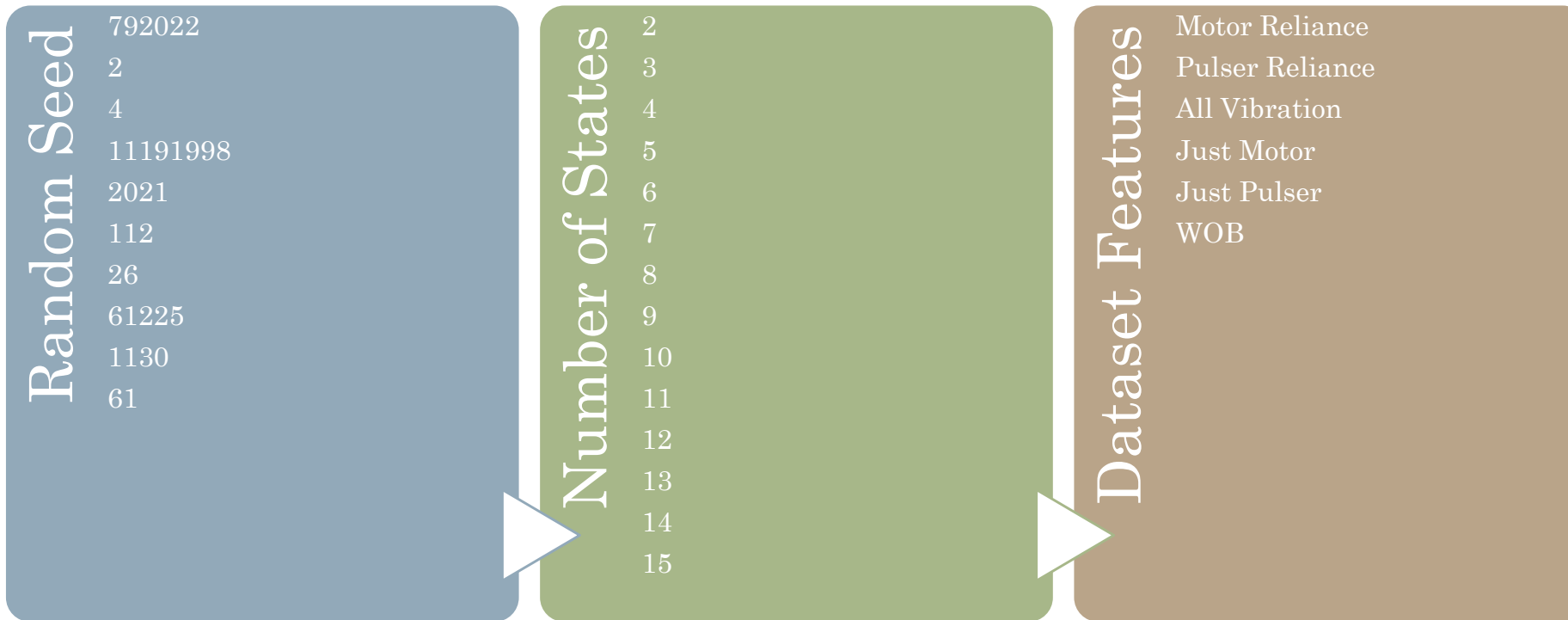
The Data

- Provided by DDC from real runs on oil rigs
- Includes the following items:
 - Time
 - Depth
 - Gallons per Minute (GPM) (Flow)
 - RPM
 - Axial and Lateral Vibration (at Motor and Pulser)
 - Weight on Bit (WOB)

Preprocessing

DRILLING PARAMETER	ORIGINAL DATA	PREPROCESSED DATA	PROCESSING CHANGES
TIME	RT_Time	Time	Rename, Fill Down
DEPTH	RT_Depth	Depth	Rename, Fill Down
FLOW	- RT_Pumps_Off - RT_Pumps_On	GPM	Aggregate, Fill Down
ROTATION	MR_RPM-AVG.MR	Motor_RPM	Rename, Fill Down
VIBRATION	- MR_VIBA, MR_VIBL - PD_Axial Vibration, PD_Lateral Vibration	- Motor_Axial_Vibration - Motor_Lateral_Vibration - Pulser_Axial_Vibration - Pulser_Lateral_Vibration	Rename, Fill Down
WOB	RT_WOB	WOB	Rename, Fill Down

Model Variations



Evaluation



**Train on Flybar 1WB
and 1WC**

5 Runs in Total



**Test on Flybar
2WC**

1 Run



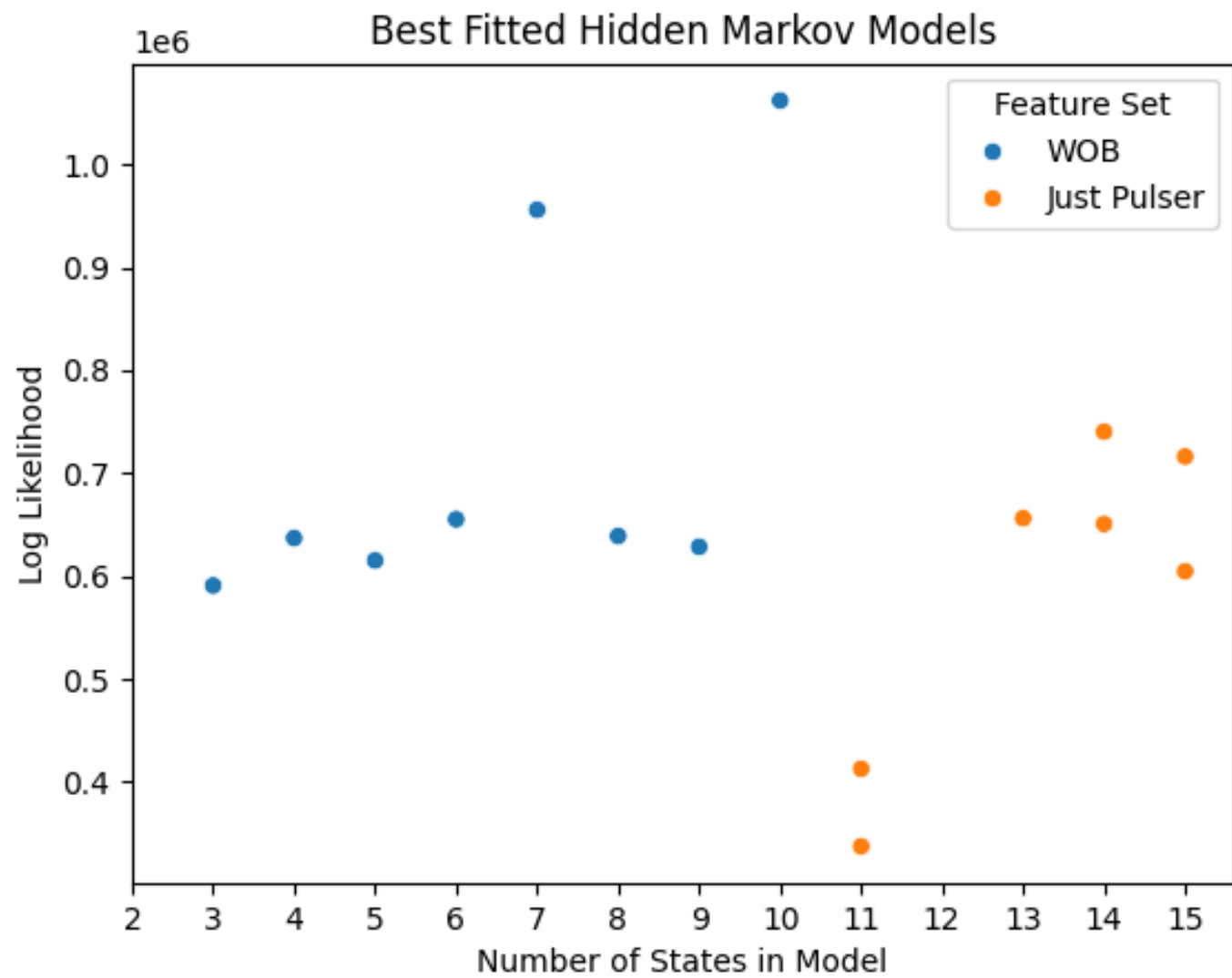
Statistical Analysis

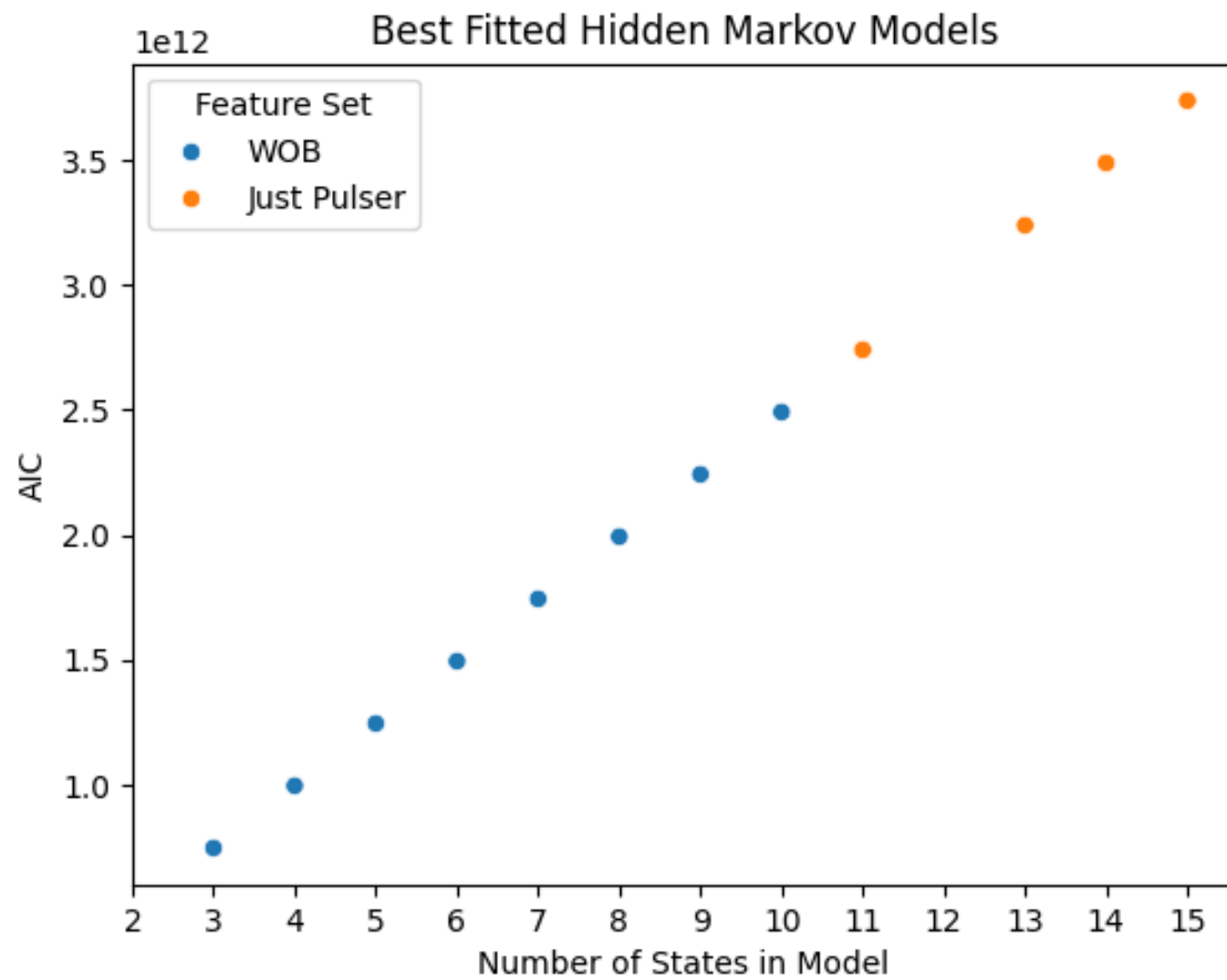
Domain Knowledge

Log Likelihood

AIC/BIC

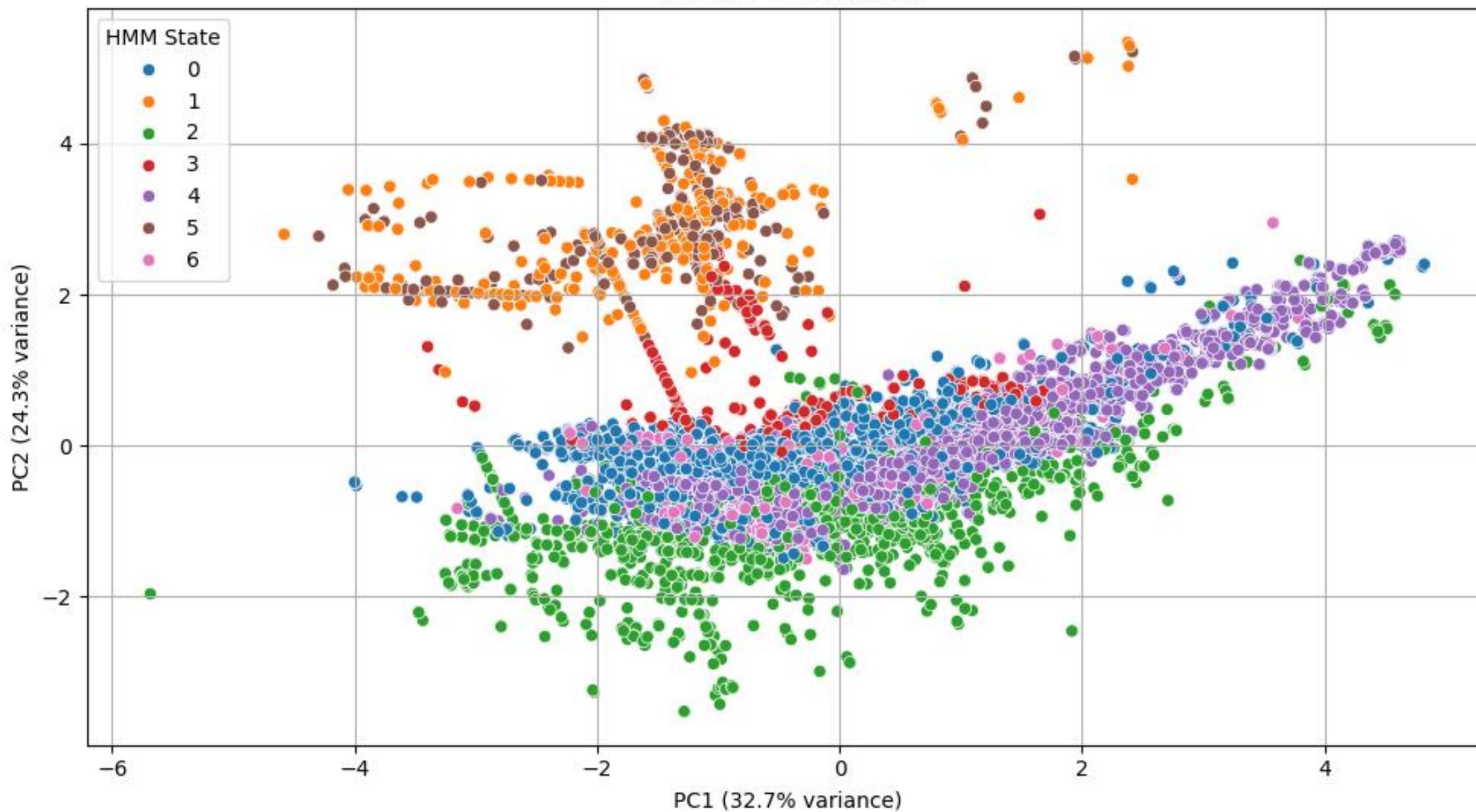
Results



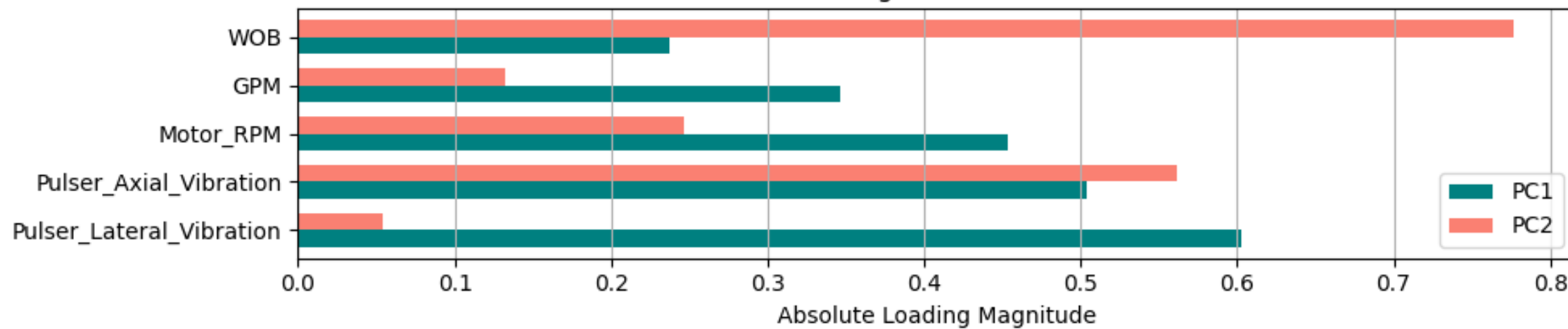


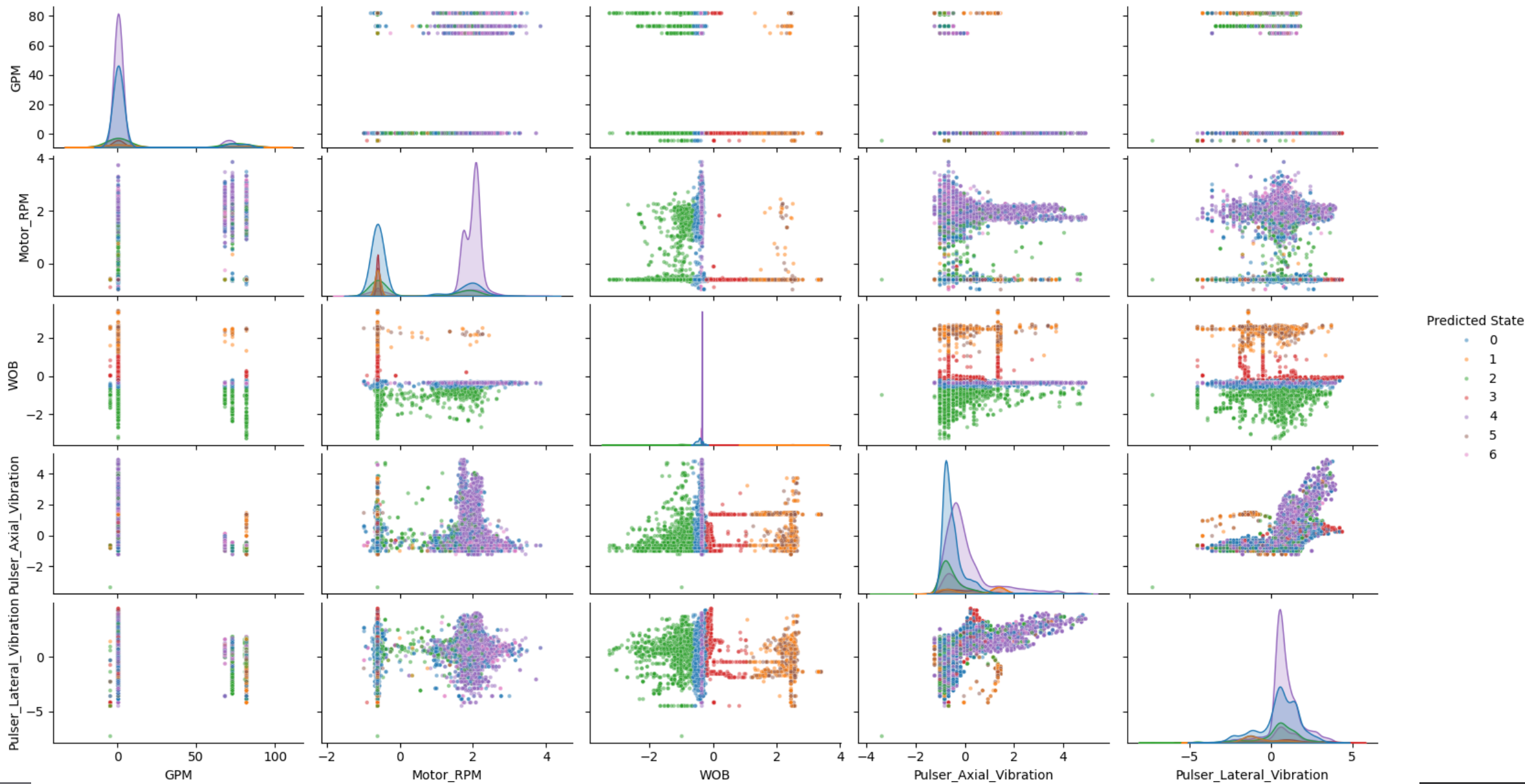
Discussion

PCA of Features Colored by Predicted HMM State
(7-State WOB Model)



Absolute Loadings of 7-State WOB Model



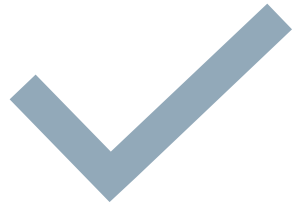


Challenges/Lessons Learned

- Initially planning to use data labels
 - Found they were unreliable
- Pivoted to HMM
 - Created well-fit models
 - States uninterpretable
- Preprocessing Obfuscated Results
 - Sparse data did not handle forward filling well



Future Work



Continued Unsupervised Approaches

Preprocess Differently

GMM

HDBSCAN



Supervised Approaches

Gather new, labeled data

RF

Direct implementation

Likely most accurate method